

作業三

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Warning: The hard deadline has passed. You can attempt it, but **you will not get credit for it**. You are welcome to try it as a learning exercise.

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Question 1

Questions 1-2 are about linear regression.

Consider a noisy target $y = \mathbf{w}_f^T \mathbf{x} + \epsilon$, where $\mathbf{x} \in \mathbb{R}^d$ (with the added coordinate $x_0 = 1$), $y \in \mathbb{R}$, $\mathbf{w}_f$ is an unknown vector, and ϵ is a noise term with zero mean and σ^2 variance.

Assume ϵ is independent of $\mathbf{x}$ and of all other ϵ 's. If linear regression is carried out using a training data set $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$, and outputs the parameter vector $\mathbf{w}_{\text{lin}}$, it can be shown that the expected in-sample error E_{in} with respect to $\mathcal{D}$ is given by:

$$\mathbb{E}_{\mathcal{D}}[E_{\text{in}}(\mathbf{w}_{\text{lin}})] = \sigma^2 \left(1 - \frac{d+1}{N}\right)$$

For $\sigma = 0.1$ and $d = 8$, which among the following choices is the smallest number of examples N that will result in an expected E_{in} greater than 0.008?

- ☐ 10
- ☐ 25
- ☐ 100
- ☐ 500

Question 2

Recall that we have introduced the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ in class, where $\mathbf{X} \in \mathbb{R}^{N \times (d+1)}$. Assume $\mathbf{X}^T \mathbf{X}$ is invertible, which statements of $\mathbf{H}$ are true?

- ☐ Some eigenvalues of $\mathbf{H}$ are bigger than 1.
- ☐ $\mathbf{H}$ is positive semi-definite.
- ☐ $\mathbf{H}$ is always invertible.
- ☐ $d + 1$ eigenvalues of $\mathbf{H}$ are 1.

Question 3

Questions 3-5 are about error and SGD

Which of the following are upper bounds of $[\text{sign}(\mathbf{w}^T \mathbf{x}) \neq y]$ for $y \in \{-1, +1\}$?

- ☐ $err(\mathbf{w}) = \max(0, 1 - y\mathbf{w}^T \mathbf{x})^2$
- ☐ $err(\mathbf{w}) = \theta(-y\mathbf{w}^T \mathbf{x})$
- ☐ $err(\mathbf{w}) = \max(0, 1 - y\mathbf{w}^T \mathbf{x})$
- ☐ $err(\mathbf{w}) = \max(0, -y\mathbf{w}^T \mathbf{x})$

Question 4

Which of the following are differentiable functions of $\mathbf{w}$ everywhere?

- ☐ $err(\mathbf{w}) = \max(0, 1 - y\mathbf{w}^T \mathbf{x})^2$
- ☐ $err(\mathbf{w}) = \theta(-y\mathbf{w}^T \mathbf{x})$
- ☐ $err(\mathbf{w}) = \max(0, -y\mathbf{w}^T \mathbf{x})$
- ☐ $err(\mathbf{w}) = \max(0, 1 - y\mathbf{w}^T \mathbf{x})$

Question 5

When using SGD on the following error functions and 'ignoring' some singular points that are not differentiable, which of the following error function results in PLA?

- ☐ $err(\mathbf{w}) = \max(0, 1 - y\mathbf{w}^T \mathbf{x})$
- ☐ $err(\mathbf{w}) = \max(0, -y\mathbf{w}^T \mathbf{x})$

- ☐ $err(\mathbf{w}) = \theta(-y\mathbf{w}^T \mathbf{x})$
- ☐ $err(\mathbf{w}) = (\max(0, 1 - y\mathbf{w}^T \mathbf{x}))^2$

Question 6

For Questions 6-10, you will play with gradient descent algorithm and variants. Consider a function

$$E(u, v) = e^u + e^{2v} + e^{uv} + u^2 - 2uv + 2v^2 - 3u - 2v.$$

What is the gradient $\nabla E(u, v)$ around $(u, v) = (0, 0)$?

- ☐ $(-3, 1)$
- ☐ $(3, -1)$
- ☐ $(0, -2)$
- ☐ $(-2, 0)$

Question 7

In class, we have taught that the update rule of the gradient descent algorithm is

$$(u_{t+1}, v_{t+1}) = (u_t, v_t) - \eta \nabla E(u_t, v_t)$$

Please start from $(u_0, v_0) = (0, 0)$, and fix $\eta = 0.01$, what is $E(u_5, v_5)$ after five updates?

- ☐ 1.436
- ☐ 2.825
- ☐ 3.277
- ☐ 4.904

Question 8

Continue from Question 7, if we approximate the $E(u + \Delta u, v + \Delta v)$ by $\hat{E}_2(\Delta u, \Delta v)$, where $\hat{E}_2$ is the second-order Taylor's expansion of E around $(u, v) = (0, 0)$. Suppose

$$\hat{E}_2(\Delta u, \Delta v) = b_{uu}(\Delta u)^2 + b_{vv}(\Delta v)^2 + b_{uv}(\Delta u)(\Delta v) + b_u \Delta u + b_v \Delta v + b.$$

What are the values of $(b_{uu}, b_{vv}, b_{uv}, b_u, b_v, b)$ around $(u, v) = (0, 0)$?

- ☐ (1.5, 4, -1, -2, 0, 3)
- ☐ (3, 8, -0.5, -1, -2, 0)
- ☐ (3, 8, -1, -2, 0, 3)
- ☐ (1.5, 4, -0.5, -1, -2, 0)

Question 9

Continue from Question 8 and denote the Hessian matrix to be $\nabla^2 E(u, v)$, and assume that the Hessian matrix is positive definite. What is the optimal $(\Delta u, \Delta v)$ to minimize $\hat{E}_2(\Delta u, \Delta v)$? The direction is called the \textit{Newton Direction}?

- ☐ $+\left(\nabla^2 E(u, v)\right)^{-1} \nabla E(u, v)$
- ☐ $-\nabla^2 E(u, v) \nabla E(u, v)$
- ☐ $-\left(\nabla^2 E(u, v)\right)^{-1} \nabla E(u, v)$
- ☐ $+\nabla^2 E(u, v) \nabla E(u, v)$

Question 10

Using the Newton direction (without η) to update, please start from $(u_0, v_0) = (0, 0)$, what is $E(u_5, v_5)$ after five updates?

- ☐ 4.532
- ☐ 1.279
- ☐ 2.361
- ☐ 3.046

Question 11

For Questions 11-12, you will play with feature transforms

Consider six inputs $\mathbf{x}_1 = (1, 1)$, $\mathbf{x}_2 = (1, -1)$, $\mathbf{x}_3 = (-1, -1)$, $\mathbf{x}_4 = (-1, 1)$, $\mathbf{x}_5 = (0, 0)$, $\mathbf{x}_6 = (1, 0)$. What is max set of those inputs can be shattered by some quadratic, linear, or constant hypotheses of $\mathbf{x}$?

- ☐ $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$
- ☐ $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4$
- ☐ $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6$
- ☐ $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5$

Question 12

Assume that a transformer peeks the data and decides the following transform Φ "intelligently" from the data of size N . The transform maps $\mathbf{x} \in \mathbb{R}^d$ to $\mathbf{z} \in \mathbb{R}^N$, where

$$(\Phi(\mathbf{x}))_n = z_n = [\mathbf{x} \cdot \mathbf{x}_n]$$

Consider a learning algorithm that performs linear classification after the feature transform, what is $d_{vc}(\mathcal{H}_\Phi)$?

- ☐ $N + 1$
- ☐ $d + 1$
- ☐ ∞
- ☐ 1

Question 13

For Questions 13-15, you will play with linear regression and feature transforms. Consider the target function:

$$f(x_1, x_2) = \text{sign}(x_1^2 + x_2^2 - 0.6)$$

Generate a training set of $N = 1000$ points on $\mathcal{X} = [-1, 1] \times [-1, 1]$ with uniform

probability of picking each $\mathbf{x} \in \mathcal{X}$. Generate simulated noise by flipping the sign of the output in a random 10% subset of the generated training set.

Carry out Linear Regression without transformation, i.e., with feature vector: $(1, x_1, x_2)$, to find the weight $\mathbf{w}$. What is the closest value to the classification in-sample error (E_{in})? Run the experiment 1000 times and take the average E_{in} in order to reduce variation in your results.

- ☐ 0.3
- ☐ 0.5
- ☐ 0.8
- ☐ 0.1

Question 14

Now, transform the training data into the following nonlinear feature vector:

$$(1, x_1, x_2, x_1 x_2, x_1^2, x_2^2)$$

Find the vector $\tilde{\mathbf{w}}$ that corresponds to the solution of Linear Regression.

Which of the following hypotheses is closest to the one you find using Linear Regression on the transformed input? Closest here means agrees the most with your hypothesis (has the most probability of agreeing on a randomly selected point).

- ☐ $g(x_1, x_2) = \text{sign}(-1 - 0.05x_1 + 0.08x_2 + 0.13x_1x_2 + 1.5x_1^2 + 15x_2^2)$
- ☐ $g(x_1, x_2) = \text{sign}(-1 - 0.05x_1 + 0.08x_2 + 0.13x_1x_2 + 15x_1^2 + 1.5x_2^2)$
- ☐ $g(x_1, x_2) = \text{sign}(-1 - 1.5x_1 + 0.08x_2 + 0.13x_1x_2 + 0.05x_1^2 + 0.05x_2^2)$
- ☐ $g(x_1, x_2) = \text{sign}(-1 - 0.05x_1 + 0.08x_2 + 0.13x_1x_2 + 1.5x_1^2 + 1.5x_2^2)$

Question 15

Following the previous question, what is the closest value to the classification out-of-sample error E_{out} of your hypothesis? (Estimate it by generating a new set of 1000 points and adding noise as before. Average over 1000 runs to reduce the variation in your results).

- ☐ 0.1
- ☐ 0.8

- ☐ 0.5
- ☐ 0.3

Question 16

For Questions 16-17, you will derive an algorithm for multinomial (multiclass) logistic regression. For a K -class classification problem, we will denote the output space $\mathcal{Y} = \{1, 2, \dots, K\}$. The hypotheses considered by MLR are indexed by a list of weight vectors $(\mathbf{w}_1, \dots, \mathbf{w}_K)$, each weight vector of length $d + 1$. Each list represents a hypothesis

$$h_y(\mathbf{x}) = \frac{\exp(\mathbf{w}_y^T \mathbf{x})}{\sum_{i=1}^K \exp(\mathbf{w}_i^T \mathbf{x})}$$

that can be used to approximate the target distribution $P(y|\mathbf{x})$. MLR then seeks for the maximum likelihood solution over all such hypotheses.

For general K , derive an $E_{\text{in}}(\mathbf{w}_1, \dots, \mathbf{w}_K)$ like page 11 of Lecture 10 slides by minimizing the negative log likelihood.

- ☐ $\frac{1}{N} \sum_{n=1}^N \left(\sum_{i=1}^K (\mathbf{w}_i^T \mathbf{x}_n - \mathbf{w}_{y_n}^T \mathbf{x}_n) \right)$
- ☐ $\frac{1}{N} \sum_{n=1}^N \left(\ln \left(\sum_{i=1}^K \exp(\mathbf{w}_i^T \mathbf{x}_n) - \exp(\mathbf{w}_{y_n}^T \mathbf{x}_n) \right) \right)$
- ☐ $\frac{1}{N} \sum_{n=1}^N \left(\sum_{i=1}^K \mathbf{w}_i^T \mathbf{x}_n - \mathbf{w}_{y_n}^T \mathbf{x}_n \right)$
- ☐ $\frac{1}{N} \sum_{n=1}^N \left(\ln \left(\sum_{i=1}^K \exp(\mathbf{w}_i^T \mathbf{x}_n) \right) - \mathbf{w}_{y_n}^T \mathbf{x}_n \right)$

Question 17

For the E_{in} derived above, its gradient ∇E_{in} can be represented by

$\left(\frac{\partial E_{\text{in}}}{\partial \mathbf{w}_1}, \frac{\partial E_{\text{in}}}{\partial \mathbf{w}_2}, \dots, \frac{\partial E_{\text{in}}}{\partial \mathbf{w}_K} \right)$, write down $\frac{\partial E_{\text{in}}}{\partial \mathbf{w}_i}$.

- ☐ $\frac{1}{N} \sum_{n=1}^N ((h_i(\mathbf{x}_n) - 1)\mathbf{x}_n)$
- ☐ $\frac{1}{N} \sum_{n=1}^N ((h_i(\mathbf{x}_n) - [y_n = i])\mathbf{x}_n)$
- ☐ $\frac{1}{N} \sum_{n=1}^N \left(\sum_{i=1}^K (\exp(\mathbf{w}_i^T \mathbf{x}_n) - [y_n = i])\mathbf{x}_n \right)$

☐ $\frac{1}{N} \sum_{n=1}^N \left(\sum_{i=1}^K (\exp(\mathbf{w}_i^T \mathbf{x}_n) - 1) \mathbf{x}_n \right)$

Question 18

For Questions 18-20, you will play with logistic regression.

Please use the following set for training:

https://d396qusza40orc.cloudfront.net/ntumlone%2Fhw3%2Fhw3_train.dat

and the following set for testing:

https://d396qusza40orc.cloudfront.net/ntumlone%2Fhw3%2Fhw3_test.dat

Implement the fixed learning rate gradient descent algorithm for logistic regression. Run the algorithm with $\eta = 0.001$ and $T = 2000$, what is $E_{out}(g)$ from your algorithm, evaluated using the 0/1 error on the test set?

- ☐ 0.322
- ☐ 0.220
- ☐ 0.475
- ☐ 0.103

Question 19

Implement the fixed learning rate gradient descent algorithm for logistic regression. Run the algorithm with $\eta = 0.01$ and $T = 2000$, what is $E_{out}(g)$ from your algorithm, evaluated using the 0/1 error on the test set?

- ☐ 0.103
- ☐ 0.322
- ☐ 0.475
- ☐ 0.220

Question 20

Implement the fixed learning rate stochastic gradient descent algorithm for logistic regression.

Instead of randomly choosing n in each iteration, please simply pick the example with the cyclic order $n = 1, 2, \dots, N, 1, 2, \dots$

Run the algorithm with $\eta = 0.001$ and $T = 2000$, what is $E_{out}(g)$ from your algorithm, evaluated using the 0/1 error on the test set?

- ☐ 0.473
- ☐ 0.328
- ☐ 0.219
- ☐ 0.105

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